2	(a)	atoms / molecules / particles behave as elastic (identical) spheres	(1)	
		volume of atoms / molecules negligible compared to volume of containing vessel	(1)	
		time of collision negligible to time between collisions	(1)	
		no forces of attraction or repulsion between atoms / molecules	(1)	
		atoms / molecules / particles are in (continuous) random motion	(1)	
		(*any four, 1 each*)	B4	[4]
	(b)	$pV = \frac{1}{3}Nm\langle c^2\rangle$ and $pV = nRT$ *or* $pV = NkT$	B1	
		$\frac{1}{3}Nm\langle c^2\rangle = nRT$ or $= NkT$ and $\langle E_K\rangle = ½m\langle c^2\rangle$	B1	
		$n = N/N_A$ *or* $k = R/N_A$	B1	
		$\langle E_K\rangle = \frac{3}{2} \times R/N_A \times T$	A0	[3]
	(c) (i)	reaction represents *either* build-up of nucleus from light nuclei *or* build-up of heavy nucleus from nuclei	M1	
		so fusion reaction	A1	[2]
	(ii)	proton and deuterium nucleus will have equal kinetic energies	B1	
		$1.2 \times 10^{-14} = \frac{3}{2} \times 8.31 / (6.02 \times 10^{23}) \times T$	C1	
		$T = 5.8 \times 10^8$ K	A1	[3]
		(*use of* $E = 2.4 \times 10^{-14}$ *giving* 1.16×10^9 *K scores 1 mark*)		
	(iii)	*either* inter-molecular / atomic / nuclear forces exist *or* proton and deuterium nucleus are positively charged / repel	B1	[1]

3 (a) (i) 8.0 cm A1 [1]